

B

1. (a) The IQR = $7.4 - 5.8 = 1.6$, so $1.5 \times \text{IQR} = 2.4$
A whisker is allowed to extend as much as $1.5 \times \text{IQR}$
- below the 1st quartile — or to $5.8 - 2.4 = 3.4$
 - above the 3rd quartile — or to $7.4 + 2.4 = 9.8$
- Since 9.1 is not beyond this (at the high end), it is not considered an outlier.
- (b) When Z-scores are used, we have identified data values with $|Z| \geq 2$ — that is, when $Z \leq -2$ or $Z \geq 2$ — as possible outliers.
- (c) The Z-score for 9.1 is $Z = \frac{9.1 - 6.591}{1.288} \doteq 1.945$
By this method, then, 9.1 is not flagged as an outlier.

2. Only options (a) and (b)

3. (a) $P(A^c) = 1 - P(A) = 1 - 0.35 = \underline{0.65}$

(b) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
 $= 0.35 + 0.4 - 0.15 = \underline{0.6}$

(c) $P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.15}{0.4} \doteq \underline{0.375}$